

SHOPIFY DATASET 2025

Data Cleaning, Exploratory Data Analysis & Time Series
Visualization

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SUBJECT: PYTHON FOR DATA ANALYSIS (WEEK 4 ASSIGNMENT)

DATASET: SHOPIFY DATASET 2025

COLLAB LINK:

<https://colab.research.google.com/drive/12yIBxLkPVKGo4bBjniHUvtrj7Q9Jpg4O?usp=sharing>

Section 1 - Objective

- The primary objective of this project is to perform an end-to-end exploratory financial analysis on Shopify's daily stock data spanning May 2015 to March 2025 (2,469 total trading days). The dataset is cleaned, transformed, and evaluated to identify price trends, volume dynamics, moving average indicators, daily percentage return distributions, and market volatility peaks.

Section 2 - Historical Price Trend

- To evaluate historical growth, a closing price line plot was generated. Shopify experienced massive long-term expansion from an initial baseline of \$1.93 up to a peak closing price of \$169.06 in late 2021, followed by market correction and post-2023 stabilization around \$75–\$100.

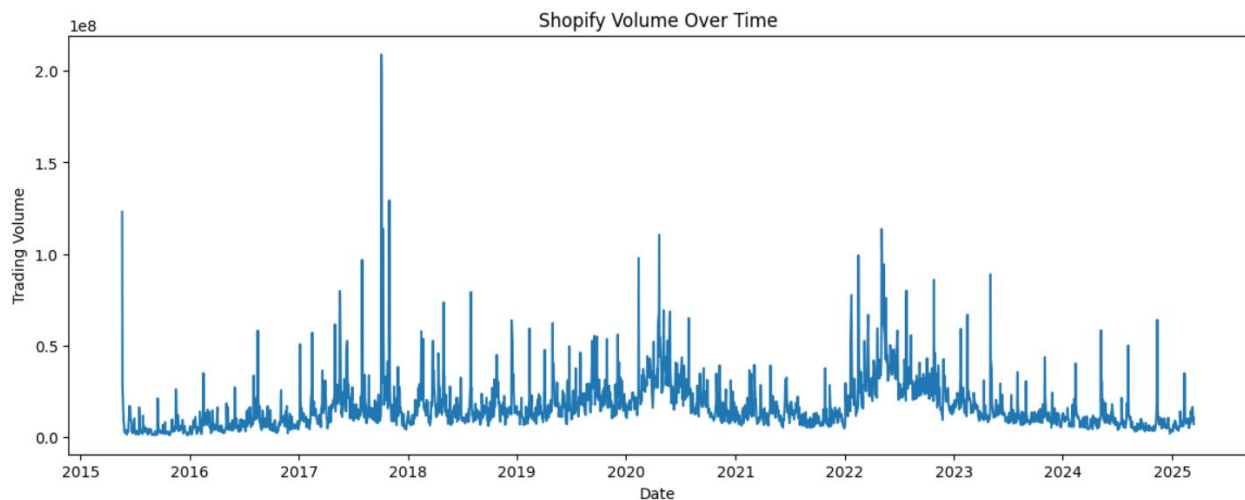
```
plt.figure(figsize=(12,5))
sns.lineplot(data=df,x=df.index,y="close")
plt.title("Shopify Closing Price Over Time")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.show()
```



Section 3 - Trading Volume Analysis

- Trading volume reflects liquidity and investor participation. The average daily volume for Shopify was 16.57 million shares, with significant volume surges during earnings announcements and major macroeconomic shifts, reaching a peak daily volume of 208.96 million shares.

```
plt.figure(figsize=(14,5))
sns.lineplot(data=df,x="date",y="volume")
plt.title("Shopify Volume Over Time")
plt.xlabel("Date")
plt.ylabel("Trading Volume")
plt.show()
```



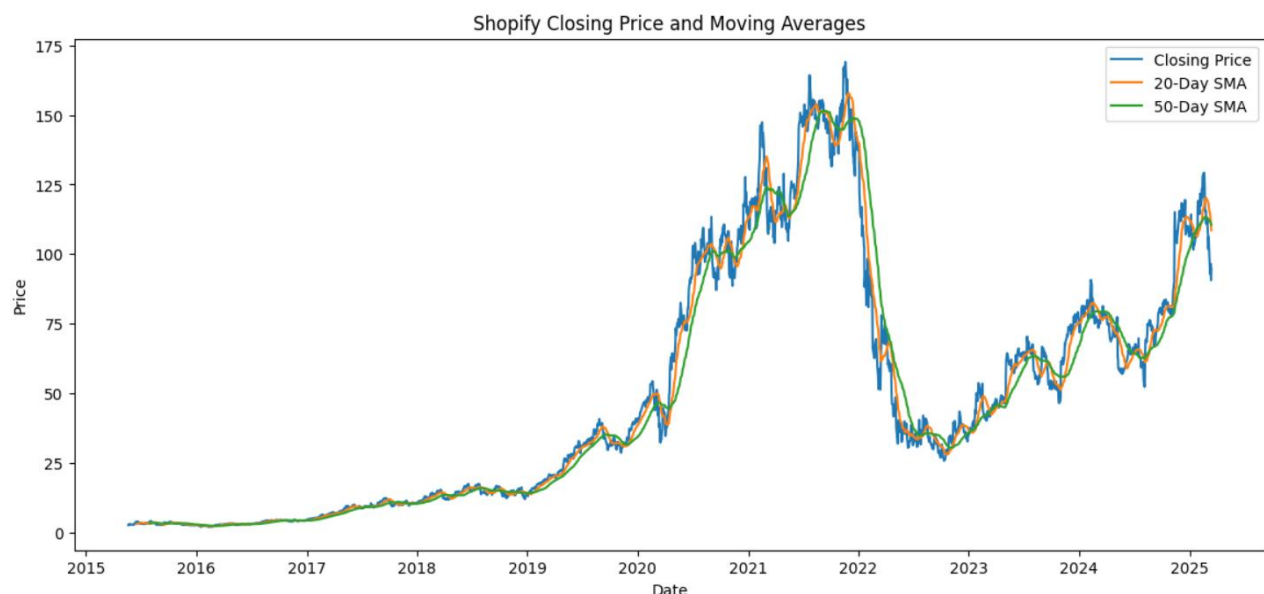
Section 4 - Moving Average Analysis

- To smooth short-term price fluctuations and identify technical trends, 20-day Simple Moving Average (20-Day SMA) and 50-day Simple Moving Average (50-Day SMA) lines were plotted alongside the daily closing price.

```

plt.figure(figsize=(14, 6))
sns.lineplot(data=df,x=df.index,y="close",label="Closing Price")
sns.lineplot(data=df, x=df.index,y="SMA_20",label="20-Day SMA")
sns.lineplot(data=df, x=df.index,y="SMA_50",label="50-Day SMA")
plt.title("Shopify Closing Price and Moving Averages")
plt.xlabel("Date")
plt.ylabel("Price")
plt.show()

```



Section 5 - Return Distribution

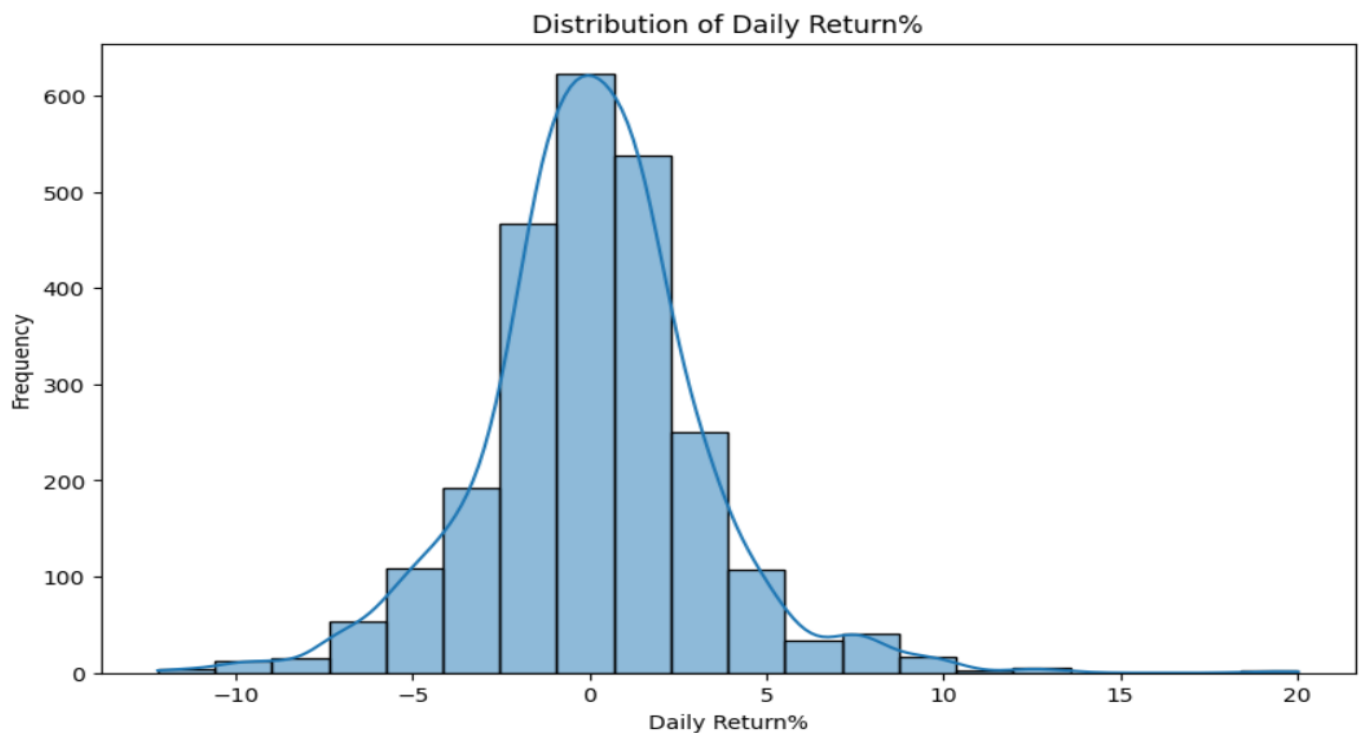
- Daily percentage returns were calculated to inspect the risk profile and distribution shape using a Histogram paired with Kernel Density Estimation (KDE). The returns display a tight bell curve centered near 0% return with heavy tails, indicating frequent low daily movement punctuated by extreme jump days.

```
plt.figure(figsize=(10,6))

sns.histplot(data=df, x="daily return%", bins=20, kde=True) # Kernel density estimate
plt.title("Distribution of Daily Return%")

plt.xlabel("Daily Return%")
plt.ylabel("Frequency")

plt.show()
```



Section 6 - Volatility Analysis

- Market risk was isolated by calculating absolute daily percentage swings. Filtering for top volatile trading dates identified key market events with price jumps or drops exceeding 15–20% in a single trading session.

volatility

```
df["Absolute Return"] = df["daily return%"].abs()
```

```
df.nlargest(10, "Absolute Return")
```

	open	high	low	close	adj_close	volume	daily price change	SMA_20	SMA_50	daily return%	Absolute Return
date											
2022-03-18 00:00:00-04:00	64.987999	78.000000	64.730003	78.000000	78.000000	46309000	13.012001	62.031950	81.609080	20.022160	20.022160
2015-09-17 00:00:00-04:00	2.989000	3.795000	2.985000	3.555000	3.555000	21123000	0.566000	2.773950	3.136220	18.936098	18.936098
2022-02-24 00:00:00-05:00	58.502998	66.878998	58.502998	66.398003	66.398003	35520000	7.895004	81.711900	105.081180	13.495042	13.495042
2022-05-12 00:00:00-04:00	31.361000	38.500000	30.806000	35.351002	35.351002	94274000	3.990002	44.737450	56.483300	12.722814	12.722814
2022-07-20 00:00:00-04:00	34.189999	38.840000	33.950001	38.490002	38.490002	42330800	4.300003	33.946850	34.738560	12.576786	12.576786
2022-01-24 00:00:00-05:00	83.400002	96.399002	78.000000	93.848999	93.848999	77556000	10.448997	117.935099	137.080059	12.528774	12.528774
2022-01-21 00:00:00-05:00	100.469002	101.363998	86.875000	88.211998	88.211998	53351000	-12.257004	120.439299	138.151079	-12.199787	12.199787
2022-07-27 00:00:00-04:00	31.459999	35.480000	31.309999	35.240002	35.240002	79941500	3.780003	33.954000	34.863540	12.015266	12.015266
2017-10-04 00:00:00-04:00	11.683000	11.693000	10.075000	10.330000	10.330000	208959000	-1.353000	11.738000	10.639100	-11.580927	11.580927
2022-04-20 00:00:00-04:00	59.298000	59.400002	52.500000	52.500000	52.500000	51187000	-6.798000	65.232800	67.115040	-11.464131	11.464131

Section 7 - Key Financial Insights

- **Significant Multi-Year Growth:** Shopify stock expanded over 8,600% from its initial minimum closing price of \$1.93 (2015) to an peak price of \$169.06 (2021).
- **Pronounced Volatility Profile:** With a daily close standard deviation of \$43.57 relative to a mean of \$48.46, Shopify exhibits substantial exposure to broader market cycles.
- **Trading Volume Liquidity Correlation:** Volume spikes strongly correlate with major earnings announcements and volatility shocks, peaking at 208.96M shares in a single session.
- **Trend Momentum via Moving Averages:** Crossovers between the 20-Day SMA and 50-Day SMA reliably signal macroeconomic trend shifts.
- **Fat-Tailed Return Distribution:** Daily percentage return analysis highlights strong kurtosis (fat tails), showing that tail-risk events (single-day swings >10%) happen far more frequently than standard normal distribution models predict.

Section 8 - Conclusion

- The Final Week 4 Report successfully establishes a clean, modular technical analysis framework for Shopify stock data. By moving beyond raw price plotting into volume trend analysis, 20/50-day SMA tracking, daily return distribution modeling (Hist + KDE), and top volatility date identification, the notebook delivers a robust quantitative foundation for future time-series forecasting models.